

Adaptive Graph Filtering Neural Network for Graph Anomaly Detection

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Abstract—Graph anomaly detection (GAD) plays an important role in improving public safety and product quality and has attracted a great deal of interest in recent years. Although a wide range of progress has been achieved recently, the following challenges still remain: (1) abnormal nodes mixed in the normal node subgraph and (2) global-consistency filtering to different features. To overcome these challenges, we propose AGFNN, a novel adaptive graph filtering neural network designed to handle diverse mixed local patterns and feature variations, thereby improving model fitting from both the node and feature perspectives. Specifically, to enhance the discriminative capacity of the node representation, channel-wise feature adaptive filtering is proposed to learn a specific filter for each feature in a progressive way, which first performs multi-frequency filtering and then adaptively captures the importance of different frequency components for each feature. Meanwhile, to better fit the complex local subgraphs, the node's preference for multi-frequency information can be self-adjusted by learning node-aware bias, which is also equal to learning a specific filter for each node. Extensive experiments on real-world graph datasets demonstrate that AGFNN outperforms the state-of-the-art methods.

Index Terms—Graph anomaly detection, graph filtering, graph representation, graph neural networks.

I. INTRODUCTION

AN ANOMALY is defined as data that deviates significantly from the other members of the sample in which it occurs [1]. It is widely present in various real scenarios such as defective products in industry [2], [3], [4] and abnormal users in finance [5]. As such, automated anomaly detection, which

is critical to improving public safety and product quality, has attracted considerable interest from both academia and industry. Beyond conventional data formats, anomaly detection is also highly valuable in the context of graph data. With the wide presence of graph-structured data in various domains, such as user review records in a social platform [6], [7] and transaction records in a financial product [8], [9], graph anomaly detection (GAD) [21] has become a very popular research topic with high practical value in the field of anomaly detection. In a graph, the entities and relationships are represented as nodes and edges. GAD usually aims to automatically detect abnormal nodes or abnormal edges with the help of rich graph information.

Prior to the remarkable advance of deep learning in the graph domain, earlier works [10], [11], [12] focused on designing node-related statistical features, such as in/out degree, based on the given graph structure information to detect abnormal nodes. Recently, due to the excellent performance of GNNs in various domains [13], [14], [15], [16], [17], a wide range of methods have explored using GNNs to learn discriminative node representations for detecting abnormal nodes. Specifically, considering the camouflage behaviors of abnormal nodes, i.e., abnormal nodes usually connect normal nodes, the spatial-based GNNs have been designed with specific neighbor sampling and aggregation functions [5], [7], [18], [19], [20] to better aggregate information from a local subgraph, thereby increasing the separability of normal and abnormal nodes. Additionally, since the homophily assumption is not available in the GAD task, recent spectral-based GNN methods [21], [22] proposed multi-frequency filters to capture both low-frequency and high-frequency signals for improving the discrepancy.

Although some encouraging results have been achieved, we believe that existing approaches still neglect the following two issues, which limit further improvement in anomaly detection performance: (i) *abnormal nodes mixed in the normal node subgraph*. When abnormal nodes appear in the graph, the homophily ratio of a graph presents a significant variation, as shown in Fig. 1. This indicates that abnormal nodes exhibit strong heterophily, which means that abnormal nodes prefer to connect with normal nodes. This structural camouflage makes abnormal nodes tend to be similar to the surrounding normal nodes under the message-passing framework. Meanwhile, it is observed that some normal nodes may also exhibit heterogeneity due to interference from a large number of anomalous nodes (e.g., the Elliptic dataset). This phenomenon suggests that prior approaches, which determine the importance of frequency components based on node categories and encourage normal nodes

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The code will be available at <https://github.com/TFboys-lzz/AGFNN>.

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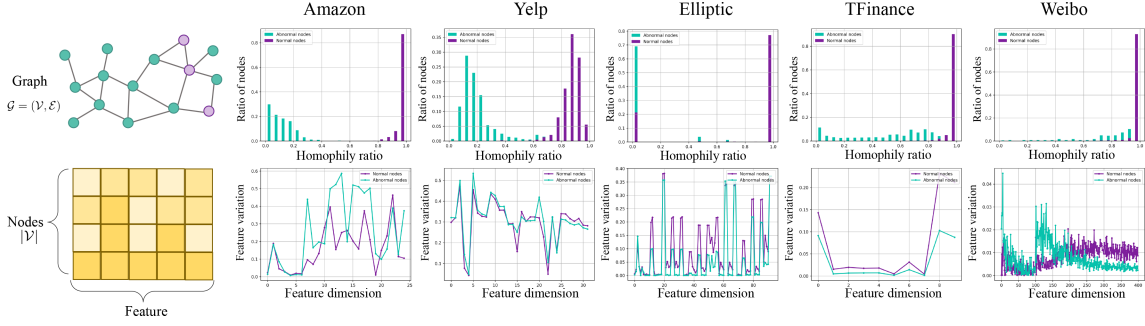


Fig. 1. The statistical information of five real-world graph datasets, including the homophily ratio $h_{N_1}(v)$ and feature variation $\mathcal{D}$ of abnormal (in green color) and normal nodes (in purple color). Obviously, due to the camouflage in both structure and feature levels, normal and abnormal nodes show significant differences in two metrics.

to focus on low-frequency signals while forcing abnormal nodes to attend to high-frequency signals [22], may not be appropriate.

In addition to the graph structure information, multi-dimensional graph signals are also provided to describe the node information, which leads to the second issue: (ii) *Global-consistency filtering across different features*. As different features capture distinct attributes of a node, using a global filter for different features may restrict the discriminative capacity of the node representation. To provide a qualitative analysis, we define a new metric (named feature variation) to calculate the feature difference between a node and its neighbors, and the metric results are shown in Fig. 1. It is evident that for both normal and abnormal nodes, the variation in features manifests differently across local subgraphs. Meanwhile, as a complementary perspective, we further observe that while some normal nodes display heterogeneity in the Elliptic dataset, the feature variation metrics reveal notable differences in local feature variation between the two types of nodes. This insight prompts us to account for such feature variations on the GAD task. Although the feature variation metric does not directly quantify the significance of specific frequency components, it highlights that applying a fixed filter across different features may fail to capture the diverse patterns of variation. Building upon this observation, we are motivated to extend the conventional graph filtering to a more generalized form, aiming to capture a more discriminative representation.

Inspired by the above observations, we are convinced that addressing these challenges effectively would greatly improve the performance of the GAD task. In particular, given the complex local subgraphs resulting from structural camouflage, it is evident that adaptively aggregating neighborhood information is crucial for enhancing the separability between normal and abnormal nodes. Meanwhile, independently adapting to different feature channels is also necessary to preserve the distinct properties of each feature channel. To this end, we aim to strengthen spectral-based GNNs for the GAD task and propose adaptive frequency filtering from both structure and feature perspectives. First, the multiple frequency information was explicitly extracted based on the designed filtering group, and the importance analysis of each frequency component is conducted for adapting channel-wise variations, which helps augmenting the discriminative ability of node representation. Then, the

adaptive frequency filtering at the structure level is introduced to dynamically aggregate specific frequency information from local subgraphs. In summary, our **main contributions** can be highlighted in the following:

- We propose an **Adaptive Graph Filtering Neural Network** for GAD (namely AGFNN), in which both the complex topological information and the inconsistent variation of node features are considered.
- To enhance the discriminative capability of the node representation, channel-wise feature adaptive filtering is proposed by multi-frequency filtering and adaptively learning the importance of different frequency components for each feature channel.
- To fit the complex local subgraphs, we also propose to learn node-aware bias for adjusting the preference of the node for different frequency information.
- Extensive experiments on a range of real-world graph datasets show that our AGFNN outperforms existing state-of-the-art methods, achieving further improvements.

II. RELATED WORKS

A. Graph Anomaly Detection

Recently, with the success of graph neural networks (GNNs) in a variety of graph learning tasks [16], [17], a large number of studies have explored using GNNs for graph anomaly detection (GAD), as shown in Table I. According to the perspective of information processing, existing GAD methods can be roughly categorized into *spatial-based* and *spectral-based* approaches. Specifically, we further categorize the existing spatial-based GAD methods into three classes.

1) *Graph reconstruction-based methods*. Among spatial-based approaches, one representative paradigm is graph reconstruction methods [7], [23], [24], [25], [26], [38], [39], [40], which aim to reconstruct the graph topology or node attributes and then detect anomalies based on reconstruction errors. This line of work assumes that abnormal nodes, due to their inconsistency with the overall graph structure, are more difficult to reconstruct than normal ones. Early efforts mainly employed autoencoder-based architectures. For instance, DOMINANT [23] designed a deep autoencoder framework that simultaneously reconstructs the node attributes and adjacency

TABLE I

SUMMARY OF RELATED WORKS ON NODE-LEVEL GRAPH ANOMALY DETECTION. FULLY-SUPERVISED INDICATES THAT THE RATIO OF TRAINING/VALIDATION/TEST DATASETS IS 40%/40%/20%, WHILE THE RATIO OF TRAINING DATA IS SET TO 1% IN THE SEMI-SUPERVISED SCENARIO.

Category	Method	Year	Supervision paradigm	Algorithm	Dataset
Graph reconstruction	DOMINANT [23]	2019	Unsupervised	Structure + attribute reconstruction	BlogCatalog, Flickr, and ACM
	HO-GAT [24]	2023	Unsupervised	Hybrid-order attention + reconstruction	Scholar, AMiner, and WebKB
	GAD-NR [25]	2024	Unsupervised	Self + neighborhood reconstruction	Cora, Weibo, Reddit, Disney, Books, and Enron
	G3AD [26]	2025	Unsupervised	Auxiliary networks + correlation constraints	Cora, Citeseer, Weibo, Reddit, Pubmed, and Flickr
Aggregation mechanism	CARE-GNN [18]	2020	Semi-supervised & fully-supervised	Label-aware neighbor selection	Yelp and Amazon
	PCGNN [27]	2021	Fully-supervised	Label-balanced sampler + subgraph construction	Yelp, Amazon, and two private datasets
	H2-FDetector [28]	2022	Semi-supervised	Homophilic/heterophilic informed aggregation	Yelp and Amazon
	PMP [29]	2024	Semi-supervised & fully-supervised	Node-specific aggregation functions	Yelp, Amazon, T-Finance, and T-Social
Graph contrastive learning	IHGAT [30]	2021	Semi-supervised & fully-supervised	Heterogeneous transaction-intention network	Private dataset
	SL-GAD [31]	2021	Unsupervised	Multi-view contrastive + generative regression	BlogCatalog, Flickr, ACM, Cora, Citeseer, and Pubmed
	GCCAD [32]	2022	Unsupervised	Link inference + contrastive learning	AMiner, MAS, Alpha, and Yelp
	ARISE [33]	2023	Unsupervised	Region proposal + substructure contrastive	Cora, Citeseer, DBLP, Citation, ACM, and Pubmed
Spectral filtering	AMNet [22]	2022	Semi-supervised	Low- and high-frequency spectral filtering	YelpChi, Elliptic, FinV, and Telecom
	BWGNN [21]	2022	Semi-supervised & fully-supervised	Beta-wavelet band-pass filter	YelpChi, Amazon, T-Finance, and T-Social
	SEC-GFD [34]	2024	Fully-supervised	Mixed-frequency hybrid filtering	YelpChi, Amazon, T-Finance, and T-Social
	CIE-GAD [35]	2025	Fully-supervised	Hypergraph + spectral convolution	YelpChi, Amazon, Elliptic, and T-Finance
	AHFAN [36]	2025	Semi-supervised	Semantic-based multiple frequency information fusion	YelpChi, Amazon, Elliptic, Weibo, and Tolokers
	FAGAD [37]	2025	Unsupervised	Frequency self-adaptation across bands	BlogCatalog, Flickr, Amazon, Enron, Reddit, Wiki, and Facebook
	AGFNN	2025	Semi-supervised & fully-supervised	Channel-wise feature adaptive filtering + node-aware aggregation	YelpChi, Amazon, Elliptic, T-Finance, Weibo, Reddit, Tolokers, and Questions

matrix, using the reconstruction error as the anomaly indicator. To further enhance the representation learning capacity, subsequent studies incorporated higher-order or community-level structural cues. HO-GAT [24] introduced a hybrid-order graph attention mechanism that jointly models node- and motif-level representations. Similarly, ComGA [38] integrated community-aware representations to handle local, global, and structural anomalies in a unified framework, while SI-HGAD [39] abstracted hierarchical substructures to capture multi-scale information for anomaly detection. Recent methods have explored reconstruction strategies tailored for complex or noisy graphs. GAD-NR [25] reconstructed not only a node's attributes but also its entire neighborhood, thus capturing both local and contextual discrepancies. To mitigate overfitting and alleviate the homophily trap, ADA-GAD [40] proposed a two-stage framework consisting of anomaly-denoised augmentation and distribution regularization. Meanwhile, G3AD [26] introduced auxiliary networks with correlation constraints to stabilize representation learning and reduce inconsistency issues.

2) *Aggregation mechanism-based methods.* Another important line of spatial-based GAD methods focuses on improving the aggregation mechanism in GNNs. These approaches [7], [18], [27], [28], [29], [30], [41] aim to refine the neighbor aggregation process so that the model can better distinguish informative neighbors and reduce the negative effects caused by abnormal or irrelevant nodes during message passing. To optimize the GNN aggregation process, CARE-GNN [18] devised a label-aware similarity measure to identify informative neighbors and employed reinforcement learning to adaptively select the optimal number of neighbors for aggregation. PCGNN [27] further introduced a label-balanced sampler to construct subgraphs for mini-batch training, ensuring balanced aggregation across classes. To handle both homophilic and heterophilic connections, H2-FDetector [28] developed an information aggregation strategy that enables homophilic edges to propagate similar information while allowing heterophilic edges to deliver distinct information. More recently, PMP [29] adopted a node-specific aggregation function to separately aggregate neighbors from different classes, further enhancing the adaptability of the aggregation process.

3) *Graph contrastive learning-based methods.* Recently, graph contrastive learning [31], [32], [33], [42], [43], [44], [45]

has been introduced into the GAD task to improve node representation by maximizing the agreement between semantically similar nodes or subgraphs while distinguishing abnormal patterns. These methods generally design contrastive objectives or sampling strategies to capture the inherent consistency of normal nodes and expose anomalies through inconsistent representations. CoLA [42] introduced a sampling strategy to construct contrastive instance pairs and exploited local structural information from the graph to learn normal patterns. Building on this idea, NLGAD [43] employed multi-scale contrastive learning to enhance the modeling of normal behavioral patterns across different structural levels. Differently, SL-GAD [31] constructed multiple contextual subgraphs (views) around a target node and integrated a generative attribute regression module with a multi-view contrastive learning framework for anomaly detection. To further refine the contrastive process, GCCAD [32] designed a GNN encoder capable of inferring and removing suspicious links during message passing, thus improving the reliability of contrastive learning. Meanwhile, ARISE [33] proposed a region proposal module to identify high-density substructures and leverage subgraph-level information for contrastive representation learning. Recently, GRADATE [44] developed a multi-view and multi-scale contrastive learning framework that performs subgraph-to-subgraph contrast to capture both global and local anomaly cues.

Spectral-based GAD methods. Different from spatial-based methods, spectral-based GAD [21], [22], [34], [35], [36], [37], [46], [47] approaches focus on analyzing graph signals in the frequency domain. Since abnormal nodes usually exhibit dissimilar features from their neighbors, such behavior corresponds to high-frequency components in the spectral space. This observation motivates researchers to design spectral filtering strategies that can effectively capture and combine information from both low- and high-frequency domains for anomaly detection. IDGL [46] designed an adaptive dual-channel convolution filter that combines low- and high-frequency information from neighboring nodes to both assimilate nodes with assortative connections and distinguish those with disassortative edges. Similarly, AMNet [22] employed a multi-frequency filter group to jointly learn representations from different spectral bands. Tang et al. [21] observed a “right-shift” phenomenon in the spectral distribution caused by anomalies and proposed a beta

wavelet-based GNN with band-pass filtering capability to address this issue. By dividing the spectrum into multiple mixed-frequency bands, Xu et al. [34] developed a hybrid filtering module to alleviate the effects of heterophily. Based on Chebyshev polynomial filtering, Wang et al. [36] proposed a semantic fusion-based node representation framework to integrate information from various frequency components. CIE-GAD [35] introduced a spectral convolution mechanism with node-level attention fusion to adaptively capture multifrequency signals. Furthermore, FAGAD [37] developed a frequency self-adaptation framework that dynamically fuses signals across multiple frequency bands to enhance representation discriminability.

B. Spectral-Based Graph Neural Networks

The main idea of spectral-based GNNs [48], [49], [50], [51], [52], [53], [54] is to design spectral graph filters in the spectral domain, utilizing the theory of graph signal processing. Chebnet [48] employed Chebyshev polynomials to approximate a filter, and in this work, GCN [55] adopted a first-order approximation to further simplify ChebNet. To tackle the over-smooth problem contained in the original GCN, Dong et al. [49] designed AdaGNN, which achieved adaptive frequency response filters for different feature channels to capture the varying importance of different frequency components. Similarly, FAGCN [56] proposed to adaptively integrate different frequency signals by a self-gating mechanism in the aggregation process. Recently, to improve the reliability of the learned filters, the BernNet [50] was proposed to estimate an arbitrary graph filter by a K-order Bernstein polynomial approximation. Furthermore, Zheng et al. [51] proposed node-oriented spectral filtering for a graph neural network to capture precise local node positioning via the generalized translated operator.

III. PRELIMINARIES

A. Notations

For a given undirected graph, it contains a set of nodes $\mathcal{V} = \{v_i\}_{i=1}^{|\mathcal{V}|}$ with $|\mathcal{V}| = n$ and a set of edges $\mathcal{E}$ among nodes, and then the graph can be denoted as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. As the most important characteristic of $\mathcal{G}$, its topological structure can be represented by the adjacency matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ where $\mathbf{A}_{i,j} = 1$ if $(i, j) \in \mathcal{E}$ or 0 otherwise. Then the diagonal degree matrix $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$ with the i -th diagonal element $d_i = \sum_j \mathbf{A}_{i,j}$ can be obtained. Meanwhile, based on the adjacency matrix $\mathbf{A}$, we can obtain the neighborhood of the node $u \in \mathcal{V}$ as $N(u)$, and the i -hop neighbors $N_i(u)$ are computed according to $N_i(u) = \{v : \text{dis}(u, v) = i \wedge v \in \mathcal{V}\}$ where $\text{dis}(u, v)$ denotes the shortest path between node u and node v on the graph $\mathcal{G}$. What's more, we can acquire the symmetric normalized Laplacian matrix of $\mathcal{G}$ by $\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}$ where $\mathbf{I}$ is the identity matrix.

In addition to the topological structure information, there is also a feature matrix $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_d] \in \mathbb{R}^{n \times d}$ about the graph signal with d channels, where $\mathbf{x}_i \in \mathbb{R}^n$ represents the i -th feature channel. For convenience, we can acquire the j -th element of $\mathbf{x}_i$ by the index operation as $\mathbf{x}_i[j]$. Specifically, we consider the

GAD task as a classification task and have a label set $\mathcal{Y} = \{0, 1\}$. For each given node $u \in \mathcal{V}$, it has a label $y_u \in \mathcal{Y}$ ($y_u = 1$ for the abnormal node and 0 otherwise).

B. Graph Fourier Transform

According to the theory of graph signal processing [57], the eigenvectors of the normalized Laplace matrix can be regarded as the basis of the graph Fourier transform. Concretely, for the Laplacian matrix $\mathbf{L}$, it can be composed by $\mathbf{L} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^\top$, where $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_n] \in \mathbb{R}^{n \times n}$ is eigenvectors matrix containing orthonormal eigenvectors $\{\mathbf{u}_l\}_{l=1}^n$, and the eigenvalues is provided in the ordered diagonal matrix $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n) \in \mathbb{R}^{n \times n}$ with $\lambda_l \in [0, 2]$, $\forall \lambda_l \in \{\lambda_l\}_{l=1}^n$. Notably, the above eigenvalues are treated as the frequencies of the graph in the spectral domain. Then, given a one-channel graph signal $\mathbf{x} \in \mathbb{R}^n$, the graph Fourier transform is defined as: $\hat{\mathbf{x}} = \mathbf{U}^\top \mathbf{x}$. The corresponding inverse graph Fourier transform is: $\mathbf{z} = \mathbf{U} \hat{\mathbf{x}}$. Thus, with a spectral filter $\hat{\mathbf{g}}$, the filter operation of the signal $\mathbf{x}$ can be denoted as:

$$\mathbf{z} = \sum_{l=1}^n \hat{g}(\lambda_l) \mathbf{u}_l \mathbf{u}_l^\top \mathbf{x} = \mathbf{U} \hat{\mathbf{g}} \mathbf{U}^\top \mathbf{x} \quad (1)$$

where $\hat{\mathbf{g}}$ is a diagonal matrix of frequency as $\hat{\mathbf{g}} = \hat{g}(\mathbf{\Lambda}) = \text{diag}(\hat{g}(\lambda_1), \dots, \hat{g}(\lambda_n)) \in \mathbb{R}^{n \times n}$ and $\mathbf{z} \in \mathbb{R}^n$ is the filtered graph signal. Further, considering that spectral filtering is equivalent to spatial convolution, we have the spatial convolutional kernel $\mathbf{g} = \mathbf{U} \hat{\mathbf{g}}$, and the spatial convolution is defined as $\mathbf{z} = \mathbf{x} *_g \mathbf{g}$ where $*_g$ denotes the graph convolution operator.

C. Polynomial-Based Spectral Filtering

In a common way, previous spectral-based methods [58] implemented the above spectral filtering by performing eigen-decomposition on the normalized Laplace matrix $\mathbf{L}$ to obtain the Fourier basis vectors $\mathbf{U}$ and the eigenvalues $\mathbf{\Lambda}$, which introduces a computational cost of $O(n^3)$. However, the expensive eigenvalue decomposition greatly limits the application of the above methods in reality. Therefore, recent works [48], [59] proposed using K-order polynomial approximation to parameterize spectral filters, without eigenvalue decomposition operations. The approximation process can be denoted as:

$$\hat{g}(\lambda_l) \approx p(\lambda_l) = \sum_{k=0}^K \theta_k \cdot b_k(\lambda_l) \quad (2)$$

where the $b_k(\cdot)$ denotes the k -th polynomial basis and θ_k is the coefficient corresponding to $b_k(\cdot)$. Further, we carry (3) to (2) and get:

$$\mathbf{z} = \mathbf{U} \hat{\mathbf{g}} \mathbf{U}^\top \mathbf{x} \approx \left(\sum_{k=0}^K \theta_k \cdot b_k(\mathbf{L}) \right) \mathbf{x} \quad (3)$$

The filtering process in (3) only involves K successive sparse matrix-vector multiplications, resulting in an overall complexity of $O(K|\mathcal{E}|)$ per forward process. Particularly, recent works have explored various polynomial basis, such as Chebyshev basis [48] and Bernstein basis [59], to accomplish the approximation of the

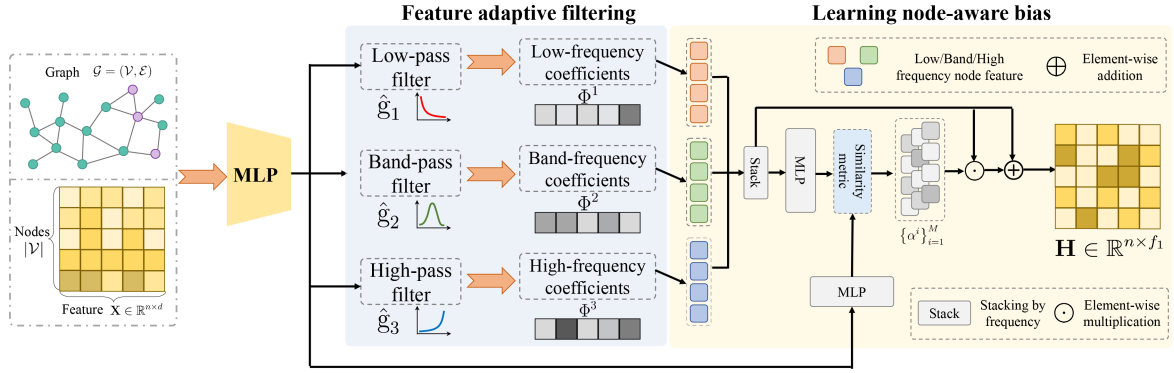


Fig. 2. Illustration of adaptive graph filtering neural network (AGFNN), which incorporates adaptive filtering from both feature and node perspectives. Specifically, the multi-frequency filter group consists of low-pass, band-pass, and high-pass filters.

graph spectral filter. As a filter with a specific property (e.g., low-pass) is fitted based on the adopted polynomial, the information corresponding to the specific frequency in the original signal can be accurately extracted.

D. Analysis of Graph Anomaly Detection

While many GNN methods have been proposed to address the camouflage behavior of abnormal nodes, the impact of abnormal nodes on the graph has not been fully quantitatively explored from the spatial domain. To this end, we first revisit the characteristics of the GAD task from both topological structure and node feature dual perspectives.

Node homophily: For the characteristics of the topological structure, we adopt the node homophily ratio of the graph as the quantitative metric. In particular, for the node $u \in \mathcal{V}$, its homophily 1-hop neighbor ratio is defined as:

$$h_{N_1(u)} = \frac{|\{v : y_u = y_v \wedge v \in N_1(u)\}|}{|N_1(u)|} \quad (4)$$

In particular, we here compute the homophily ratios of abnormal nodes $\mathcal{H}^{ano}$ and normal $\mathcal{H}^{normal}$ nodes, respectively, and for instance, $\mathcal{H}^{ano}$ is computed as:

$$\mathcal{H}^{ano} = \frac{1}{|\mathcal{V}^{ano}|} \sum_{u \in \mathcal{V}^{ano}} h_{N_1(u)} \quad (5)$$

where $\mathcal{V}^{ano}$ is the set of abnormal nodes. Following (5), the homophily ratios of normal nodes $\mathcal{H}^{normal}$ can also be obtained. As we can see in Fig. 1, the homophily ratios of normal and abnormal nodes present a relatively large difference. Therefore, for the spectral-based graph anomaly detection method, to increase the discriminative capability of the node representation, it is necessary to adaptively adjust the importance of different frequency components in the node representation according to the local subgraphs.

Feature variation: In addition to topological structures, graph data often contains multi-dimensional graph signals. One of our concerns is that using a global filter for different features may limit the discriminative ability of node representations. Therefore, a new metric $\mathcal{D} \in \mathbb{R}^d$, named feature variation, is designed to qualitatively measure the divergence. The i -th feature $\mathcal{D}[i]$ is

computed as:

$$\mathcal{D}[i] = \frac{1}{|\mathcal{V}|} \sum_{u \in \mathcal{V}} \frac{\sum_{v \in N_1(u)} \text{abs}(x_i[v] - x_i[u])}{|N_1(u)|} \quad (6)$$

where $\text{abs}(\cdot)$ indicates the absolute value operation. According to (6), a larger $\mathcal{D}[i]$ means that the i -th dimensional signal of a graph has a higher frequency. And the metric results are shown in Fig. 1. It can be clearly seen that the variation between the different features is inconsistent, indicating that different features contain various frequency information. This finding motivates us to consider that applying a global filter for different features is not the optimal solution.

IV. OVERALL FRAMEWORK.

The overall framework of our proposed adaptive graph filtering neural network (AGFNN) is shown in Fig. 2. As illustrated, given the original node features $\mathbf{X}$, a non-linear projection $\text{MLP}(\cdot)$ is first applied to $\mathbf{X}$. Then, the obtained representation $\tilde{\mathbf{X}}$ is fed to the designed multi-frequency filter group $\mathcal{F} = \{\hat{g}_i\}_{i=1}^M$ for extracting low-pass, band-pass, and high-pass frequency components. Meanwhile, the learnable coefficients $\{\Phi^i\}_{i=1}^M$ are utilized to capture the importance of different frequency components for each feature. Finally, the node representation is acquired by integrating the multi-frequency signals with the learned node-aware bias.

A. Feature Adaptive Filtering.

As we observed previously, different features present inconsistent variations within a subgraph, and as a result, the global-consistency filtering to different features may greatly limit the discriminative capability of the node representation. To tackle this issue, a natural solution would be to learn a specific filter directly for arbitrary features. However, it is not trivial as it would require expensive eigenvalue decomposition. As a preliminary attempt, a novel channel-wise feature adaptive filtering (CFAF) module is proposed to achieve the above goal in a progressive way. CFAF first performs multi-frequency filtering and then adaptively captures the importance of different frequency components for each channel. Therefore, a specific

filter for each feature is acquired by the adaptive combination of different frequency components.

To be specific, the multi-frequency filtering is achieved by a multi-frequency filter group $\mathcal{F} = \{\hat{\mathbf{g}}_i\}_{i=1}^M$ containing M graph filters which are shared to different features. And each filter $\hat{\mathbf{g}}_i \in \mathcal{F}$ possesses an excellent spectral property (e.g., low-pass) for extracting the specific frequency information. Furthermore, each feature channel can acquire multi-frequency information based on the designed filter group. Meanwhile, the proposed method should be able to adaptively adjust the importance of different frequency components based on the observed topological information and node features. To this end, when performing spectral filtering, we specify a learnable coefficient ϕ for each dimension of the node features to adaptively capture the importance of the frequency components to the feature. For example, given the single-channel graph signal $\mathbf{x}$, the filter operation in (2) can be rewritten as $\mathbf{z} = \mathbf{U}\hat{\mathbf{g}}\mathbf{U}^\top \mathbf{x}\phi$. Then, we can generalize it to the multi-frequency filter with multi-channel input $\mathbf{X} \in \mathbb{R}^{n \times d}$. More specifically, for a given graph filter $\hat{\mathbf{g}}_i \in \mathcal{F}$, we can extract signals $\mathbf{Z}^i \in \mathbb{R}^{n \times f_1}$ with specific frequency by the filter $\hat{\mathbf{g}}_i$. And the filter operation can be achieved as:

$$\mathbf{Z}^i = \mathbf{U}\hat{\mathbf{g}}_i\mathbf{U}^\top \tilde{\mathbf{X}}\Phi^i \quad (7)$$

where $\tilde{\mathbf{X}} \in \mathbb{R}^{n \times f_1} = \text{MLP}(\mathbf{X})$ denotes the projected node representation. $\Phi^i = \text{diag}(\phi_1^i, \dots, \phi_{f_1}^i) \in \mathbb{R}^{f_1 \times f_1}$ is the diagonal matrix of coefficients, and ϕ_j^i is the learnable coefficient for the j -th feature channel in i -th spectral filter. Thus, the response of each feature to the i -th frequency components can be determined by Φ^i . Furthermore, based on the learned multi-frequency parameter $\{\Phi^i\}_{i=1}^M$, the CFAF can be achieved by the combination of multiple filters. It can be found that the graph filtering generalized form in (7) degrades to the conventional form in (2) when $\Phi^i = \mathbf{I}$. In addition, as one major component of CFAF, the shared multi-frequency filter group should cover excellent spectral properties (e.g., low/band-pass filters), which can help to better estimate the specific filter for each feature. And we will introduce an available solution based on a K-order Bernstein polynomial approximation and provide detailed information about the filter design in Section IV-C.

B. Learning Node-Aware Bias.

Based on the designed multi-frequency filter group $\mathcal{F}$, the node-level representations $\mathcal{Z} = \{\mathbf{Z}^i\}_{i=1}^M$ containing different frequency information are extracted. For a given node $v \in \mathcal{V}$, the node representation can be obtained by enforcing a simple aggregation function (e.g., summation or concatenation) on the above multi-frequency signals. However, in the anomaly detection task, the importance of different frequency components to a node is influenced by its local patterns. It is predictable that the simple aggregation mechanism described above limits the capability of the model to fit complex local subgraphs. Therefore, we propose to learn node-aware bias $\{\alpha^i\}_{i=1}^M$ to adjust the importance of different frequency signals to the node representation adaptively, where $\alpha^i \in \mathbb{R}^n$ is the importance of the representation $\mathbf{Z}^i$ to all nodes $\mathcal{V}$.

Meanwhile, the key to the node-aware bias is to acquire the nodes' preferences for different frequency local information. A simple but effective solution is to compare the feature of the node itself with the extracted multi-frequency signals for learning the frequency bias. Specifically, the original node features and the extracted multi-frequency features are first mapped to a shared representation space as:

$$\begin{cases} \hat{\mathbf{Z}}^i = \text{MLP}(\mathbf{Z}^i), \forall \mathbf{Z}^i \in \mathcal{Z} \\ \hat{\mathbf{X}} = \text{MLP}(\tilde{\mathbf{X}}) \end{cases} \quad (8)$$

where $\hat{\mathbf{Z}}^i \in \mathbb{R}^{n \times f_2}$ and $\hat{\mathbf{X}} \in \mathbb{R}^{n \times f_2}$. f_2 denotes the dimension of the shared representation space. The preference of nodes $\mathcal{V}$ to the i -th frequency component $\hat{\mathbf{Z}}^i$ is computed as:

$$\mathbf{S}^i = \text{Norm}(\hat{\mathbf{Z}}^i)^\top \text{Norm}(\hat{\mathbf{X}}) \quad (9)$$

where $\mathbf{S}^i \in \mathbb{R}^{n \times n}$ is the obtained similarity matrix and $\text{Norm}(\cdot)$ denotes the row normalization operation. Specially, we focus on the node's preference for its own local multi-frequency information for the final aggregation operation. The node-aware bias α^i is acquired by extracting the diagonal elements of $\mathbf{S}^i$ as $\alpha^i = [S_{1,1}^i, S_{2,2}^i, \dots, S_{n,n}^i]$. For the j -th node v_j , the bias is normalized as $\sum_{i=1}^M \alpha^i[j] = 1$ using a softmax function. Finally, based on the learned bias $\{\alpha^i\}_{i=1}^M$, all node representations $\mathbf{H} \in \mathbb{R}^{n \times f_1}$ can be obtained as:

$$\mathbf{H} = \sum_{i=1}^M (\mathbf{I} + \mathcal{B}^i) \mathbf{Z}^i \quad (10)$$

where $\mathcal{B}^i = \text{diag}(\alpha^i) \in \mathbb{R}^{n \times n}$ is the diagonal matrix constructed from bias α^i . Obviously, the adaptive graph filtering from both node and feature dual perspectives is achieved. Furthermore, we feed the aggregated representation $\mathbf{H}$ to a classifier with the softmax function to predict the category probability of nodes. The model is optimized by the cross-entropy loss as:

$$\mathcal{L} = \frac{1}{|\mathcal{V}^{tr}|} \sum_{v_j \in \mathcal{V}^{tr}} -\log p_j[y_{v_j}] \quad (11)$$

where $p_j \in \mathbb{R}^2$ is the predicted probability of node v_j . $\mathcal{V}^{tr}$ is the set of all training nodes.

C. Construction of the Multi-Frequency Filter Group.

As previously discussed, our objective is to implement adaptive filtering for features in a progressive manner. This can be achieved by combining the multi-frequency filter group to estimate the channel-specific filter. To this end, the filter group $\mathcal{F}$ needs to encompass a range of high-quality spectral property filters. This diverse set of filters aids in reducing the complexity associated with estimating the specific filter for each channel. Specifically, to reduce the computational overhead, we adopt a simple but elegant way where the filter group is composed of a low-pass filter, a band-pass filter, and a high-pass filter, i.e., $M = 3$. Specifically, each of the above filters plays a complementary role in detecting distinct types of anomalies. The low-pass filter is particularly effective for detecting global structural anomalies, such as nodes that deviate significantly from the overall network patterns. In contrast, the band-pass filter excels at identifying

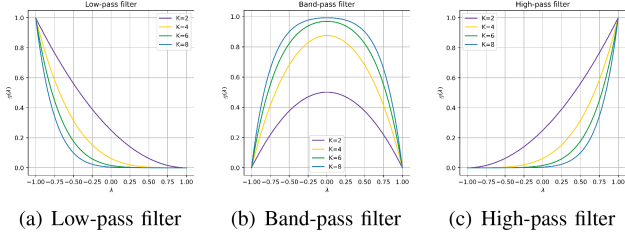


Fig. 3. The responses of low-pass, band-pass, and high-pass filters under different order K .

such subtle anomalies, where structural deviations are nuanced and localized. Meanwhile, the high-pass filter is particularly effective in enhancing the distinction between anomalous nodes with strong heterogeneity and normal nodes. Collectively, this complementary filtering strategy enables a more comprehensive representation of diverse anomaly patterns, thereby facilitating more effective anomaly detection.

Subsequently, the specific property filter for each feature, such as band-rejection and low-band-pass filters, can be obtained through adaptive combination techniques. To obtain the designed filter group, one straightforward way is to estimate the corresponding filters via K -order polynomial approximation trainable weight [48], [60]. However, those learning-based approaches are not as effective for obtaining filters with the above specific properties, since the properties of the learned filters depend on the observed graph structure and node features. Fortunately, the coefficients of the Bernstein polynomial are highly correlated with the spectral property of the target filter, which provides useful guidance when designing filters. With the help of its excellent properties, we can manually design the corresponding coefficients to approximate the filter with specific properties. We first present the definition of the Bernstein polynomial approximation below.

Definition 1 ([59]). (Bernstein polynomial approximation): For an arbitrary continuous function $h(\lambda)$ on $\lambda \in [0, 1]$, the K -order Bernstein polynomial approximation is defined as:

$$p(\lambda) = \sum_{k=0}^K \theta_k \cdot b_k^K(\lambda) = \sum_{k=0}^K \theta_k \binom{K}{k} (1-\lambda)^{K-k} \lambda^k \quad (12)$$

where $b_k^K(\lambda) = \binom{K}{k} (1-\lambda)^{K-k} \lambda^k$ is the k -th Bernstein base. θ_k is the polynomial coefficient of the k -th Bernstein basis.

Based on the above definition, the associated polynomial coefficients $\{\theta_k\}_{k=0}^K$ are manually designed to accurately approximate the shared multi-frequency group as follows:

- *Low-pass filter:* We set $\theta_k = 1$ if $k = 0$ and 0 otherwise, which yields the final approximation $p(\lambda) = (1-\lambda)^K$.
- *Band-pass filters:* We set $\theta_k = 1, \forall k \in \{1, \dots, K-1\}$, $\theta_0 = 0$ and $\theta_K = 0$, and obtain $p(\lambda) = \sum_{k=1}^{K-1} \binom{K}{k} (1-\lambda)^{K-k} \lambda^k$.
- *High-pass filter:* We set $\theta_k = 1$ if $k = K$ and 0 otherwise, and then the final approximation is $p(\lambda) = \lambda^K$.

The visualization of the approximated filters under different order K is shown in Fig. 3. It can be clearly seen that the designed coefficients provide excellent approximations to

TABLE II
THE STATISTICAL INFORMATION OF EIGHT DATASETS

Dataset	# Nodes	# Edges	Anomaly(%)	# Features
YelpChi [6]	45,954	3,846,979	14.53	32
Amazon [63]	11,944	4,398,392	6.87	25
Elliptic [8]	46,564	73,238	9.76	93
T-Finance [21]	39,357	21,222,543	4.58	10
Weibo [64]	8,405	407,963	10.3	400
Reddit [65]	10,984	168,016	3.3	64
Tolokers [66]	11,758	519,000	21.8	10
Questions [66]	48,921	153,540	3.0	301

target filters, which builds a favorable foundation for the subsequent implementation of channel-wise feature adaptive filtering. Meanwhile, the low-pass and high-pass filters exhibit steeper transitions from passband to stopband as K increases, while the band-pass filter presents a more concentrated response in the mid-frequency range. Additionally, according to the approximation theory [61], [62], for sufficiently smooth functions, the Bernstein polynomial approximation error decreases proportionally to $\mathcal{O}(1/\sqrt{K})$ as the polynomial order K increases. This implies that the approximation errors are sufficiently small to support stable and reliable channel-wise adaptive filtering in AGFNN.

V. EXPERIMENTS.

A. Experimental Setup

Datasets and baselines: Eight benchmark GAD datasets are adopted for conducting extensive experiments, including YelpChi [6], Amazon [63], Elliptic [8], T-Finance [21], Weibo [64], Reddit [65], Tolokers [66], and Questions [66]. The statistical information of all datasets is presented in Table II. We compare AGFNN with three categories of various methods: 1) four classical GNN methods, including **GCN** [55], **GAT** [67], **GIN** [68] and **GraphSAGE** [69]; 2) two representative spectral filtering methods, including **ChebNet** [48] and **BernNet** [50]. 3) four representative graph-based anomaly detection methods, including spectral filtering-based **AMNet** [22], **BWGNN** [21], data augmentation-based **ConsisGAD** [70], and message passing-based **PMP** [29]. Meanwhile, two widely adopted metrics are selected to evaluate the performance of all methods, including **F1-macro** and **AUROC**. The F1-macro is an unweighted average of the F1 scores of the normal and abnormal categories and thus can be independent of the unbalanced ratio between the two classes.

Implementation details: To comprehensively evaluate our method, both the supervised scenario (40%) and the semi-supervised scenario (1%) settings are adopted in our experiments, while the remaining data are divided into the validation and test datasets in the ratio of 1:2. On the Elliptic, Amazon, Weibo, Reddit, Questions, and Tolokers datasets, the hidden dimensions f_1 and f_2 are set to 64. For the T-Finance and YelpChi (1%) datasets, the hidden dimensions f_1 and f_2 are set to 32, while in the YelpChi (40%) dataset, they are set to 128. The polynomial order K is set to 4 for all datasets, with the exception of the elliptic dataset, where K is increased to 6. For two multi-relational YelpChi and Amazon datasets, we

TABLE III
RESULTS OF AGFNN AND COMPARED METHODS ON EIGHT DATASETS WITH 40% TRAINING RATIOS

Metric	Method	YelpChi	Amazon	T-Fin.	Ellip.	Weibo	Reddit	Tolo.	Quest.
AUROC	GCN [55]	57.11±2.16	80.29±1.14	92.13±1.11	97.05±0.27	98.49±0.62	65.07±4.08	73.71±0.74	69.95±2.46
	GAT [67]	61.98±2.66	78.28±1.86	76.38±7.65	97.53±0.30	92.10±1.97	67.98±1.90	75.44±0.57	67.12±2.44
	GIN [68]	78.20±0.65	91.42±0.66	93.38±1.66	97.80±0.11	93.61±0.38	63.72±2.94	76.65±0.69	70.96±3.08
	GraphSAGE [69]	56.26±0.39	77.08±1.87	90.69±2.45	97.10±0.22	96.23±0.70	65.94±1.13	71.70±0.85	69.63±1.03
	ChebNet [48]	82.12±2.81	92.63±2.01	89.56±4.84	97.21±0.20	97.02±0.73	68.30±7.15	82.79±0.70	71.89±1.29
	BernNet [50]	84.19±1.19	92.56±4.17	94.00±0.69	97.74±0.23	95.68±0.64	66.16±5.08	80.94±0.69	67.16±5.18
	AMNet [22]	83.54±0.52	96.11±1.96	93.85±0.75	97.63±0.18	91.59±2.20	67.58±1.30	73.81±0.96	67.11±3.61
	BWGNN [21]	82.85±0.74	95.83±2.05	95.83±0.78	98.05±0.18	95.97±1.11	66.49±4.49	79.51±0.47	68.35±4.70
	ConsisGAD [70]	77.07±1.27	95.71±1.02	96.18±0.40	95.87±0.40	95.68±0.86	67.11±2.50	75.72±1.06	71.08±0.63
	PMP [29]	82.90±2.99	97.09±0.43	96.08±0.79	95.58±0.54	95.44±0.63	55.56±1.12	75.19±0.86	66.25±5.75
	AGFNN	84.76±1.14	97.57±0.37	96.63±0.31	98.80±0.20	96.19±0.92	70.80±2.11	82.03±0.64	71.98±1.68
F1-macro	GCN [55]	56.10±1.61	66.72±1.81	86.45±0.57	91.15±0.66	94.85±0.63	51.36±1.45	62.27±0.87	58.93±0.84
	GAT [67]	61.98±2.66	64.64±1.30	71.85±7.17	91.66±0.44	89.18±1.83	52.04±1.68	63.80±0.52	56.93±0.65
	GIN [68]	66.81±0.70	91.42±0.66	89.95±1.73	92.05±0.28	92.77±0.55	53.75±0.62	63.50±0.51	59.42±1.07
	GraphSAGE [69]	64.46±1.03	64.11±1.62	85.70±0.84	89.72±0.39	93.51±0.86	51.26±0.92	60.16±0.38	57.53±0.86
	ChebNet [48]	69.93±2.91	89.89±0.79	84.10±4.92	91.91±0.47	94.51±0.41	54.85±1.07	69.73±0.78	59.63±0.90
	BernNet [50]	71.60±1.05	91.49±0.52	86.76±1.46	92.85±0.48	94.03±0.84	54.80±1.73	68.32±0.94	60.33±1.22
	AMNet [22]	71.14±0.58	91.72±4.52	88.89±1.29	92.11±0.23	89.04±1.06	54.15±1.31	61.98±0.94	60.71±1.19
	BWGNN [21]	69.95±0.66	91.43±0.79	88.57±2.55	93.81±0.18	92.75±0.72	53.23±1.40	66.84±0.47	60.07±0.72
	ConsisGAD [70]	65.03±0.87	91.46±0.70	91.00±0.56	90.14±0.49	88.54±1.24	52.86±1.67	63.08±1.07	56.87±0.72
	PMP [29]	67.22±7.27	87.04±12.34	91.35±0.79	78.53±20.22	90.72±0.88	49.30±0.66	57.45±6.95	53.69±4.20
	AGFNN	72.32±1.34	91.86±0.44	92.13±0.48	95.08±0.24	94.54±0.66	55.01±1.32	69.84±0.81	60.15±1.23

TABLE IV
RESULTS OF AGFNN AND COMPARED METHODS ON EIGHT DATASETS WITH 1% TRAINING RATIOS

Metric	Method	YelpChi	Amazon	T-Fin.	Ellip.	Weibo	Reddit	Tolo.	Quest.
AUROC	GCN [55]	55.31±2.66	71.23±5.12	90.36±1.21	79.58±0.32	89.00±6.04	57.76±4.49	69.90±0.65	57.61±6.39
	GAT [67]	58.46±3.79	79.58±0.32	67.66±4.63	90.30±1.36	64.50±8.46	58.86±3.03	58.86±3.03	58.06±4.32
	GIN [68]	64.36±3.31	87.11±3.54	93.38±1.66	91.49±1.68	66.77±5.41	54.27±4.55	70.19±1.51	60.88±5.19
	GraphSAGE [69]	55.31±2.66	75.52±3.98	86.61±2.60	92.62±1.03	71.86±9.74	58.21±3.42	69.97±0.76	62.74±2.54
	ChebNet [48]	69.21±2.58	84.36±3.01	88.88±1.29	90.82±0.61	87.17±4.05	55.10±4.67	71.54±2.03	52.46±6.94
	BernNet [50]	72.43±1.23	86.90±1.60	91.22±1.29	92.55±0.98	64.60±9.95	56.61±6.49	71.13±2.48	59.50±3.37
	AMNet [22]	72.53±0.52	86.77±7.16	92.35±1.39	92.53±1.71	67.80±13.89	58.07±3.92	68.83±1.44	56.01±3.70
	BWGNN [21]	71.68±1.40	90.06±1.78	92.22±1.37	92.08±0.93	88.16±6.87	57.10±4.09	72.23±1.53	58.55±4.18
	ConsisGAD [70]	75.28±1.12	93.04±0.73	93.44±2.43	91.61±1.66	76.65±6.12	56.52±1.61	70.21±1.94	62.65±4.72
	PMP [29]	52.44±0.27	87.70±2.86	90.39±2.16	68.65±10.52	70.91±7.98	44.65±1.50	62.58±10.11	54.35±2.59
	AGFNN	71.55±2.33	93.27±1.47	96.63±0.31	93.13±0.96	89.61±5.04	58.17±2.83	71.75±1.74	62.96±3.40
F1-macro	GCN [55]	53.90±1.59	64.89±1.18	85.43±0.61	84.76±1.18	88.64±2.47	51.69±1.28	60.41±0.87	54.10±0.99
	GAT [67]	54.81±1.47	60.81±3.43	62.57±6.54	76.91±3.31	70.28±3.58	51.58±1.12	58.30±1.11	52.52±1.33
	GIN [68]	59.41±1.32	87.11±3.54	86.34±2.76	83.42±1.22	77.75±3.03	51.34±0.97	60.32±0.67	55.07±0.99
	GraphSAGE [69]	53.90±1.59	64.76±1.92	84.34±1.42	83.77±1.21	75.53±4.15	51.31±1.30	59.76±0.63	53.08±1.46
	ChebNet [48]	60.93±1.93	81.93±4.73	81.87±1.26	81.77±1.35	82.99±3.82	52.08±1.47	61.88±1.37	53.53±1.74
	BernNet [50]	61.65±0.76	90.42±1.35	83.93±2.56	87.40±1.13	74.57±4.62	51.87±1.65	61.43±1.59	55.07±1.38
	AMNet [22]	61.88±0.81	89.12±2.10	86.76±1.76	87.36±1.76	74.68±3.60	51.60±0.89	59.52±0.96	54.89±2.23
	BWGNN [21]	61.44±0.82	90.35±0.55	86.61±1.17	85.21±1.75	82.58±4.08	51.83±1.50	61.57±0.99	55.17±1.53
	ConsisGAD [70]	61.26±1.09	90.34±0.47	88.94±1.30	86.10±1.03	71.14±4.29	49.87±0.70	59.87±1.69	55.45±1.60
	PMP [29]	36.12±15.34	84.54±4.94	84.06±11.68	47.34±0.21	62.12±9.88	49.09±1.20	46.87±10.85	50.37±2.59
	AGFNN	61.84±1.47	90.68±0.36	89.45±0.99	88.44±1.38	84.51±2.72	52.23±1.27	61.99±1.11	55.54±1.42

treat all types of edges as the same. All comparison methods are reproduced under two settings according to the official code. We report the mean and standard deviation of 10 independent runs performed with different random seeds.

B. Performance Comparison

The experimental results on eight datasets are presented in Tables III and IV, with 40% and 1% settings. As we can see, in terms of the AUROC score, AGFNN outperforms all other methods in most datasets and achieves the highest performance, such as YelpChi, Amazon, and Question datasets with 40% training data. This impressive performance improvement demonstrates that the proposed AGFNN is capable of collaboratively enhancing the discrimination of node features at both topology and representation levels, thus improving the separability of normal and abnormal nodes. More specifically, compared with traditional graph neural network methods, AGFNN adaptively extracts different frequency information based on spectral filtering, alleviating the challenge of structural camouflage to neighborhood aggregation operations. We also observe that the performance of the proposed AGFNN on the Weibo dataset is relatively lower compared to some baseline methods. This is

primarily because both normal and abnormal nodes exhibit a relatively high degree of homophily, as illustrated in Figure 1, which aligns well with the inductive bias of general GNN-based methods (i.e., low-pass filtering). Additionally, AGFNN demonstrates substantial performance gains compared with existing SOTA GAD methods, and the F1-macro improvement under the challenging 1% setting can go up to 1.18%, 0.78%, and 3.0% on YelpChi, T-Finance, and Tolokers, respectively. This is mainly due to the fact that the manually constructed multi-frequency filter group is better adapted to complex local subgraph patterns and help extract specific frequency information. Meanwhile, channel-adaptive feature filtering can adaptively capture the importance of different frequency components for each channel feature, thus improving the discriminative ability of node representations.

C. Time and Space Complexity Analysis.

To provide a clear understanding of the computational complexity in AGFNN, we present a more detailed theoretical analysis. Specifically, we decompose the proposed method into four core sub-modules, as shown in Table VII. Overall, the time complexity of AGFNN is $\mathcal{O}(Ln f_1^2 + Km f_1 +$

TABLE V
ABLATION STUDIES OF CFAF MODULE

Dataset	W/o CFAF		AGFNN	
	F1-macro	AUC	F1-macro	AUC
YelpChi (1%)	61.00±1.35	70.75±2.09	61.87±1.47	71.55±2.33
Amazon (1%)	89.73±1.23	88.99±2.39	90.48±0.67	90.20±3.00
Elliptic (1%)	88.35±1.35	92.66±1.02	88.44±1.38	93.13±0.96
T-Finance (1%)	87.60±2.40	92.29±1.08	89.45±0.99	93.57±1.36
YelpChi (40%)	70.01±1.26	82.51±1.31	72.32±1.34	84.76±1.14
Amazon (40%)	91.26±0.65	97.22±0.46	91.86±0.44	97.57±0.37
Elliptic (40%)	93.94±0.20	97.85±0.20	94.60±0.27	98.07±0.21
T-Finance (40%)	91.04±0.69	96.09±0.45	92.13±0.48	96.63±0.31

TABLE VI
ABLATION STUDIES OF EACH FREQUENCY COMPONENT

Signal	Reddit		Tolokers	
	AUC-ROC	F1-macro	AUC-ROC	F1-macro
Low frequency	66.99±1.93	52.83±0.92	79.87±0.62	66.29±0.77
Band frequency	65.66±2.17	54.71±1.28	79.74±0.59	67.70±0.66
High frequency	69.47±2.39	53.07±0.72	76.71±0.77	65.07±0.97
AGFNN	70.80±2.11	55.01±0.86	82.03±0.64	69.84±0.81

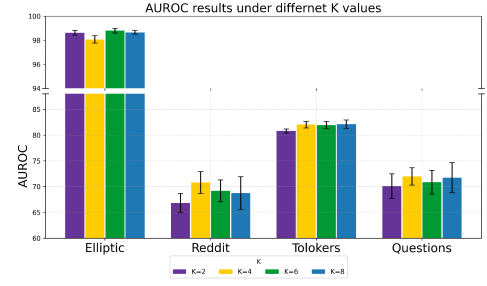
TABLE VII
TIME AND SPACE COMPLEXITY ANALYSIS OF AGFNN

Sub-modules	Feature projection	Filter approximation	Feature-adaptive filtering	Node-aware bias
Time complexity	$\mathcal{O}(Ln f_1^2)$	$\mathcal{O}(K m f_1)$	$\mathcal{O}(K n f_1)$	$\mathcal{O}(n f_1^2)$
Space complexity	$\mathcal{O}(L \cdot (n f_1 + f_1^2))$	$\mathcal{O}(K n f_1)$	$\mathcal{O}(M n f_1)$	$\mathcal{O}(f_1^2)$

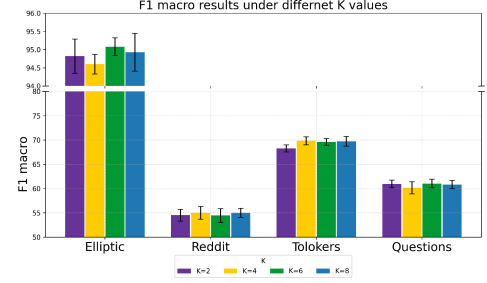
$K n f_1 + n f_1^2$), and the space complexity is $\mathcal{O}(L(n f_1 + f_1^2) + K n f_1 + M n f_1 + f_1^2)$, where we set $f_1 = f_2$. As an intuitive comparison, the time and space complexity [71] of GCN is $\mathcal{O}(L n f_1^2 + L m f_1)$ and $\mathcal{O}(L(n f_1 + f_1^2))$, respectively. It can be found that the primary additional overhead of AGFNN arises from the node-aware bias module, which introduces an extra $\mathcal{O}(n f_1^2)$ computational cost. Nevertheless, the overall complexity of AGFNN remains scalable with respect to graph size, particularly when the hidden dimension f_1 is moderate. Taking the Reddit dataset as an example, the number of total learnable parameters is 0.0212M when the hidden dimension is 64. Despite the slight increase in computation, AGFNN achieves consistent and substantial performance improvements over GCN on most benchmark datasets, demonstrating the effectiveness of the proposed adaptive graph filtering.

D. Ablation Study

1) *Effectiveness of Channel-Wise Feature Adaptive Filtering*: To evaluate the effectiveness of channel-wise feature adaptive filtering, Table V presents ablation experiment results on the four datasets. After removing the learnable parameter matrix $\{\Phi^i\}_{i=1}^M$, the model reduces to applying a global filter for different features. As we can see, after introducing the proposed CFAF, the performance of detecting abnormal nodes on four datasets has been improved, achieving 1.85% and 2.31% improvements in F1-macro on T-Finance (1%) and YelpChi (40%) datasets, respectively. These remarkable improvements demonstrate again the necessity of applying specific filters to different features. More importantly, it also proves the effectiveness of the proposed CFAF to achieve this goal in a progressive way



(a) AUROC results



(b) F1-macro results

Fig. 4. The impact of approximation order K.

through multi-frequency filtering and learning the importance of different frequency components.

2) *Effectiveness of the Adaptive Integration of Each Frequency Component*: The experimental results on the Reddit and Tolokers datasets under the 40% setting are presented in Table VI. As we can see, different frequency components contribute distinctly to various datasets, and relying on only a single frequency component tends to reduce the model's fitness. As a result, distinguishing between normal and abnormal nodes may be much less effective. In contrast, AGFNN adaptively integrates multiple frequency components, leveraging their complementary advantages to enhance the discriminative ability of node representation, thereby achieving superior anomaly detection performance.

3) *Impact of Approximation Order K*: We conduct the parameter analysis experiments to analyze the impact of the Bernstein polynomial order $K \in \{2, 4, 6, 8\}$ on the performance of AGFNN. As shown in Fig. 4, we report both AUC-ROC and F1-Macro results along with their standard deviations. It can be found that increasing K generally leads to performance improvements, especially on more complex datasets such as Reddit, Tolokers, and Yelp. For instance, on the Tolokers dataset, the F1-Macro score increases from 68.26% at $K = 2$ to 69.84% at $K = 4$, demonstrating that when the filter exhibits stronger frequency selectivity, it becomes more effective at enhancing the separability between normal and anomalous nodes. Meanwhile, we notice that on most datasets, the performance tends to stabilize when $K = 4$, suggesting that the model has reached sufficient expressive power to capture diverse anomalous patterns in a lower computational cost during the filtering process. More importantly, the choice of K can be dynamically adjusted according to the characteristics of the dataset. This flexibility is

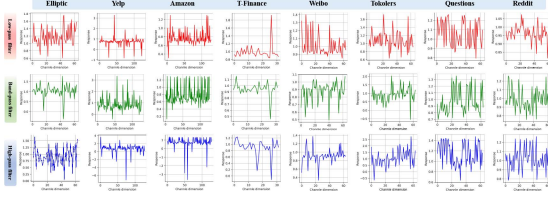


Fig. 5. The visualization of learned parameters $\{\Phi^i\}_{i=1}^M$.

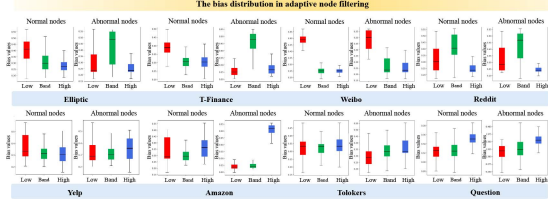


Fig. 6. The visualization of bias distribution on eight datasets.

able to balance accuracy and efficiency depending on the specific anomaly patterns and structural complexity present in the data.

E. Visualization of Feature Adaptive Filters

In addition to the quantitative experiments, we also visualized the learned coefficients matrix, i.e., $\{\Phi^i\}_{i=1}^M$, and the results are shown in Fig. 5. It can be obviously found that the learned filters present a phenomenon in which different features show various responses to multi-frequency signals. For example, some features tend to be low-pass filters, i.e., $\phi_1^i \approx 0$, while all-pass filters are more useful for some features, i.e., $\phi_j^i \neq 0, \forall j \in \{1, \dots, M\}$. According to qualitative results, the conclusion can be obtained that each feature can adaptively adjust the importance of different frequency components based on its own characteristics. Thus, the discriminative capacity of the node representation can be significantly improved.

F. Distribution of Node-Aware Bias

We also visualize the distribution of node-aware bias for normal and abnormal nodes in Fig. 6. It can be seen that normal nodes show a relatively strong preference for low-frequency information, but in some datasets, we find that high-frequency information is also important, such as in YelpChi and Amazon. This is due to the fact that homophilic local subgraphs centered at normal nodes may mix a small number of abnormal nodes. For the abnormal nodes, it can be found that the preferences present a significant right shift, with the band-frequency information and high-frequency acquiring a larger bias. Clearly, it is related to the strong heterogeneity that abnormal nodes possess. Thus, learning node-aware bias can help to adaptively integrate discriminative information from multi-frequency signals to better fit the diverse local patterns compared to other simple aggregation functions.

VI. CONCLUSION

In this paper, we first rethink the remaining challenges in graph anomaly detection, including abnormal nodes mixed in the normal node subgraph and global-consistency filtering across

different features. To overcome the above challenges, we propose a novel AGFNN that achieves adaptive graph filtering from node and feature dual perspectives for better detecting abnormal nodes. To improve the discriminative capacity of node representation, the channel-wise feature adaptive filtering is proposed to learn a specific filter for each feature in a progressive way. Meanwhile, the preference of the node for multi-frequency information is adaptively adjusted by learning node-aware bias, thus improving the fitting ability of the model to diverse local patterns. Our ongoing research work will focus on how to better tackle the multi-relation existing in the GAD task and incorporate statistical priors of node homophily into anomaly detection, thus improving the performance of detecting abnormal nodes.

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